

These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**. The purpose of this material is to make the topic clear and useful for students of Class 9 Computer Science.

Computational Thinking

Computational thinking is a problem-solving approach that involves solving problems step by step like a computer. It uses four main techniques: Decomposition, Pattern Recognition, Abstraction, and Algorithm Design to solve problems in many fields.

Examples of Computational Thinking

Example 1: Making Tea ☕

- Boil water
- Add tea leaves
- Add sugar
- Pour into cup

→ This step-by-step process is computational thinking

Example 2: Solving a Math Problem ➗

- Read the question
- Identify numbers
- Choose formula
- Solve step by step
- Write answer

→ This step-by-step process is computational thinking

Example 3: Getting Ready for School 🎒

- Wake up
- Brush teeth
- Wear uniform

- Pack bag
- Go to school

→ This step-by-step process is computational thinking

Decomposition

Decomposition means breaking a big and difficult problem into smaller, easier parts. This makes the problem easier to solve step by step.

Example: Building a Birdhouse

- Design the birdhouse (decide shape and size)
- Gather materials (wood, nails, tools)
- Cut the wood
- Assemble the parts
- Paint and decorate
- Install the birdhouse

→ So, a big task becomes simple when divided into small steps.

Note: You can even use any example that are written in Computational Thinking Topic.

Pattern Recognition

Pattern recognition means finding similarities or repeated patterns in problems.

Example1: 🙌 While studying a book: 📖

- You notice important questions repeat in exams.
- Some topics are similar in different chapters.

→ Recognizing these similarities is pattern recognition

Example2: Math Problems ➗

- Many questions follow the same formula and you apply the same method repeatedly.

Website: salmanblogs.com | Youtube: ComputerWithSalman

Example3: Even and Odd Numbers

- Sequence: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

You notice a pattern:

- Odd numbers → 1, 3, 5, 7, 9
- Even numbers → 2, 4, 6, 8, 10

Recognizing that there are 5 odd and 5 even numbers is finding a pattern

Abstraction

Abstraction means focusing only on important details and ignoring unnecessary ones. It helps reduce confusion and makes problems easier to understand.

Example 1: Studying for Exams

- Focus only on important points and formulas.
- Ignore unnecessary details like long stories or extra examples.

→ This is abstraction

Example 2: Using a TV Remote

- You press buttons to change channels or volume
- You don't need to know how the electronics inside work

→ This is abstraction

Algorithm

An algorithm is a step-by-step set of instructions to solve a problem. It's like a recipe that tells exactly what to do and in what order.

Example: Studying a Book for Exams

- Divide book into chapters

- Study one chapter at a time

Website: salmanblogs.com | Youtube: ComputerWithSalman

- Make notes of important points
- Revise all chapters
- Solve practice questions

→ Step-by-step plan = algorithm

Note: You can even use any example mentioned in Computational Thinking.

Principles of Computational Thinking

The principles of computational thinking guide us on what steps to take before actually solving a problem.

1. Problem Understanding

Before trying to solve a problem, you must fully understand it. This means identifying the main issue, the requirements, and the goals.

Importance [Advantages Of Understanding Problem]

- Clarity and Focus: Understanding the problem clearly helps you focus on what is really important and avoid distractions.
- Defining Goals: When you understand the problem, you can define clear, achievable objectives. You know what the final solution should achieve.
- Efficient Solutions: A clear understanding allows you to choose the best methods and tools, saving time and resources.
- Avoiding Mistakes: Misunderstanding the problem can lead to wrong solutions and wasted effort.

2. Problem Simplification (Decomposition)

Large or complex problems can be difficult to solve in one go. Simplifying the problem means breaking it into smaller, easier-to-handle parts (this is also called decomposition).

Example: Making Maggi 🍜

- Boil water

- Add noodles
- Add masala
- Cook
- Serve

→ Instead of one big task, you turn it into small, easy parts.

→ This breaking into steps is called decomposition.

Example: Studying a Complete Book

- Divide book into chapters
- Study one chapter at a time
- Break each chapter into topics
- Make notes for each topic
- Revise step by step

→ Instead of one big task, you turn it into small, easy parts.

→ This breaking into steps is called decomposition.

Note: Dear Students, Again you can even use other examples as well mentioned in Computational Thinking topic.

3. Solution Selection and Design

When a problem has more than one possible solution, you choose the best one and plan how to do it step by step.

Example 1: Preparing for Exams

Problem: You need to finish a book before the exam.

Possible solutions:

- Study all chapters in one day
- Study chapter by chapter over a week.

Best Solution: Study chapter by chapter → easier, less stressful, more effective.

Website: salmanblogs.com | Youtube: ComputerWithSalman

Example 2: Traveling to School 🚌

Problem: You need to reach school on time.

Possible solutions:

- Walk
- Take a bus
- Ride a bicycle

Best Solution: Take a bus → faster and less tiring.

Website: salmanblogs.com | Youtube: ComputerWithSalman